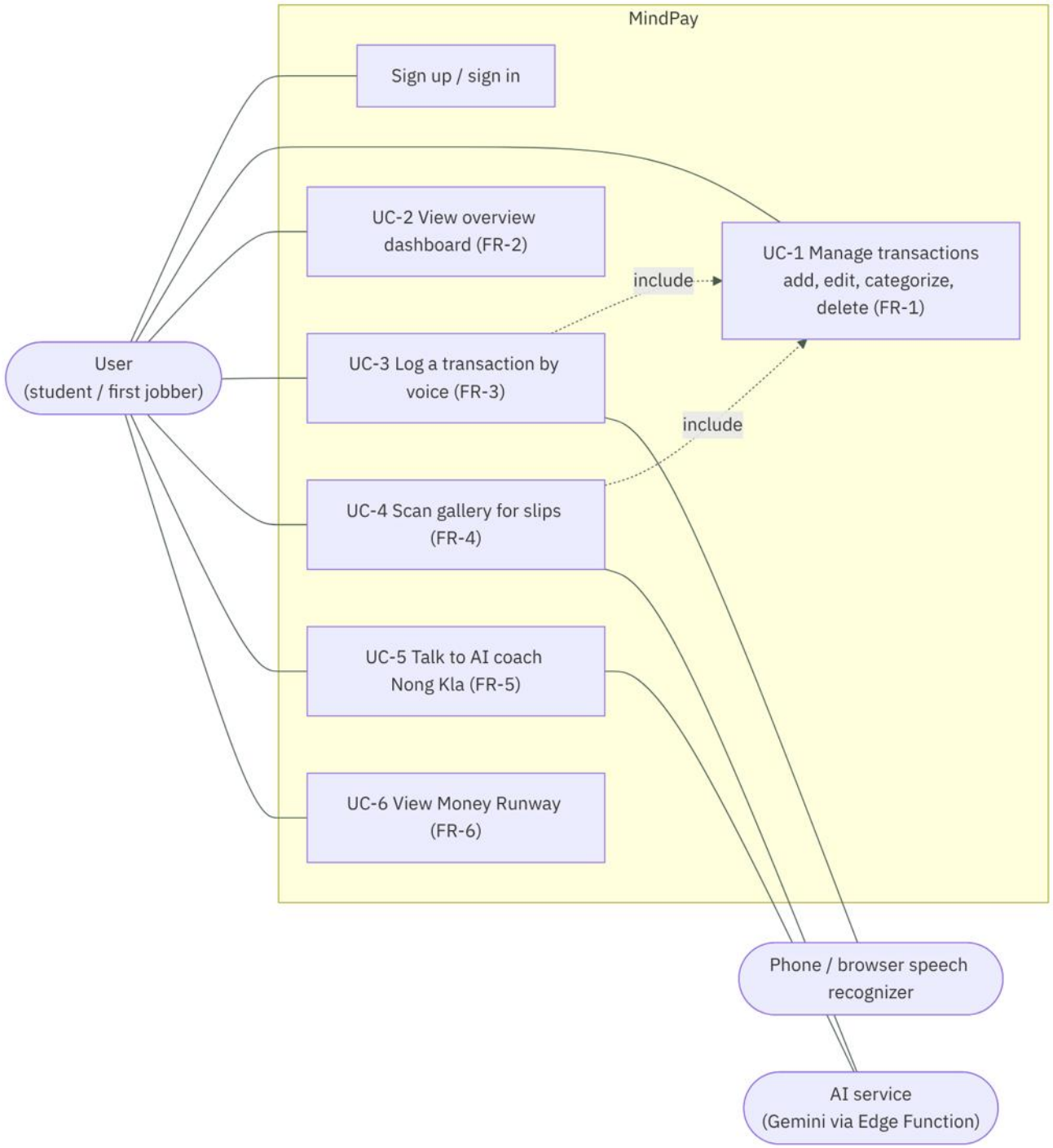
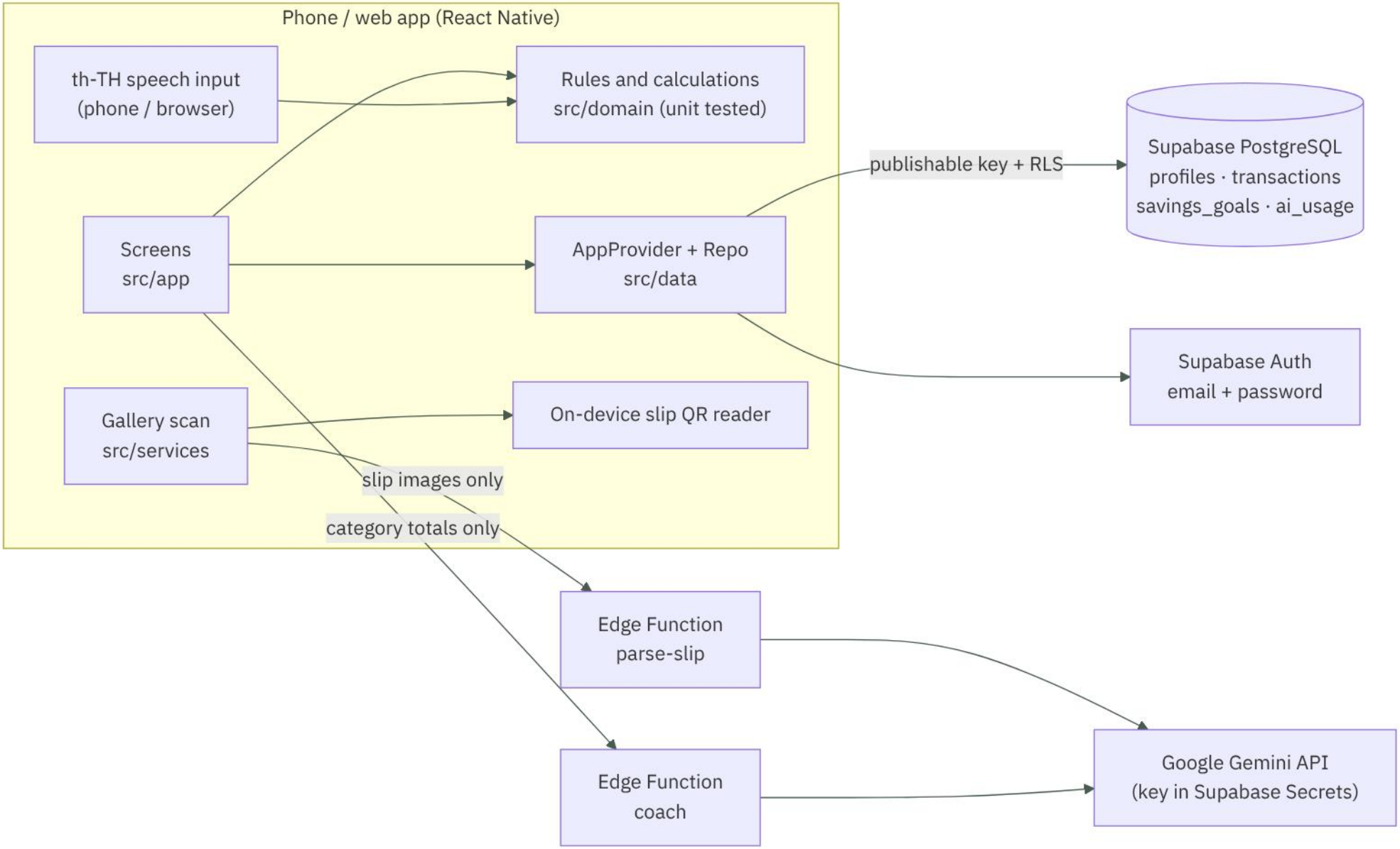
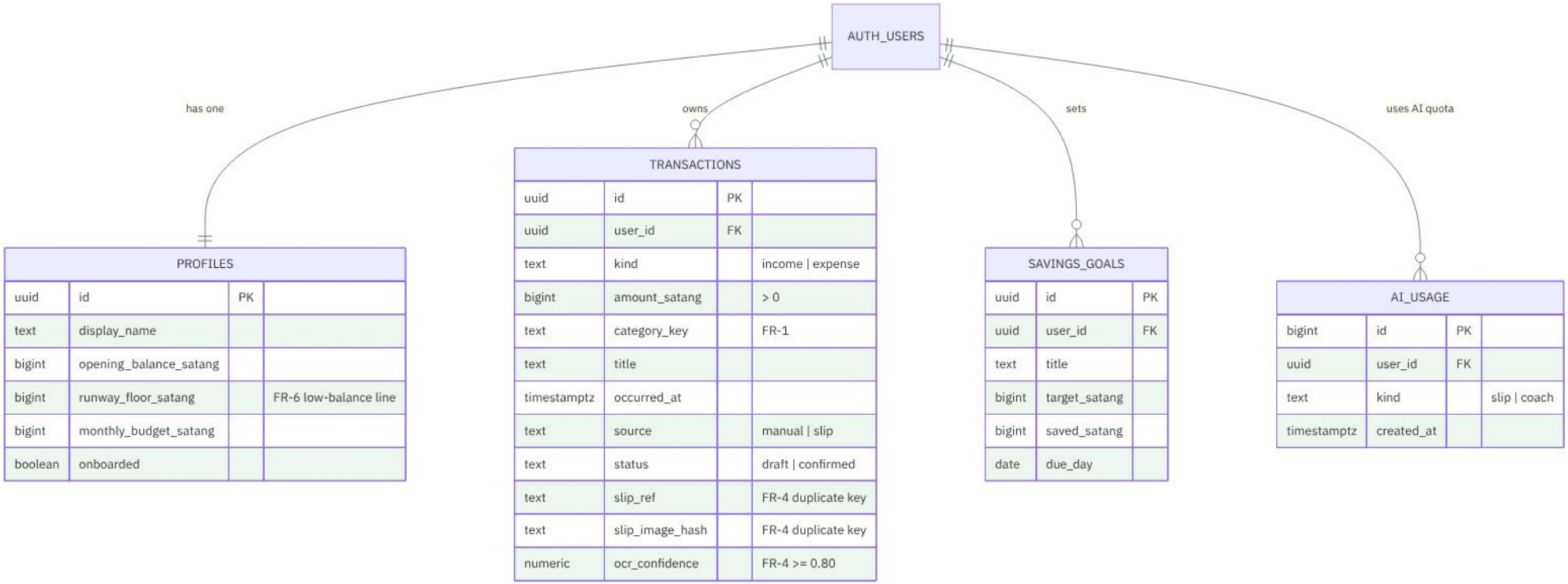
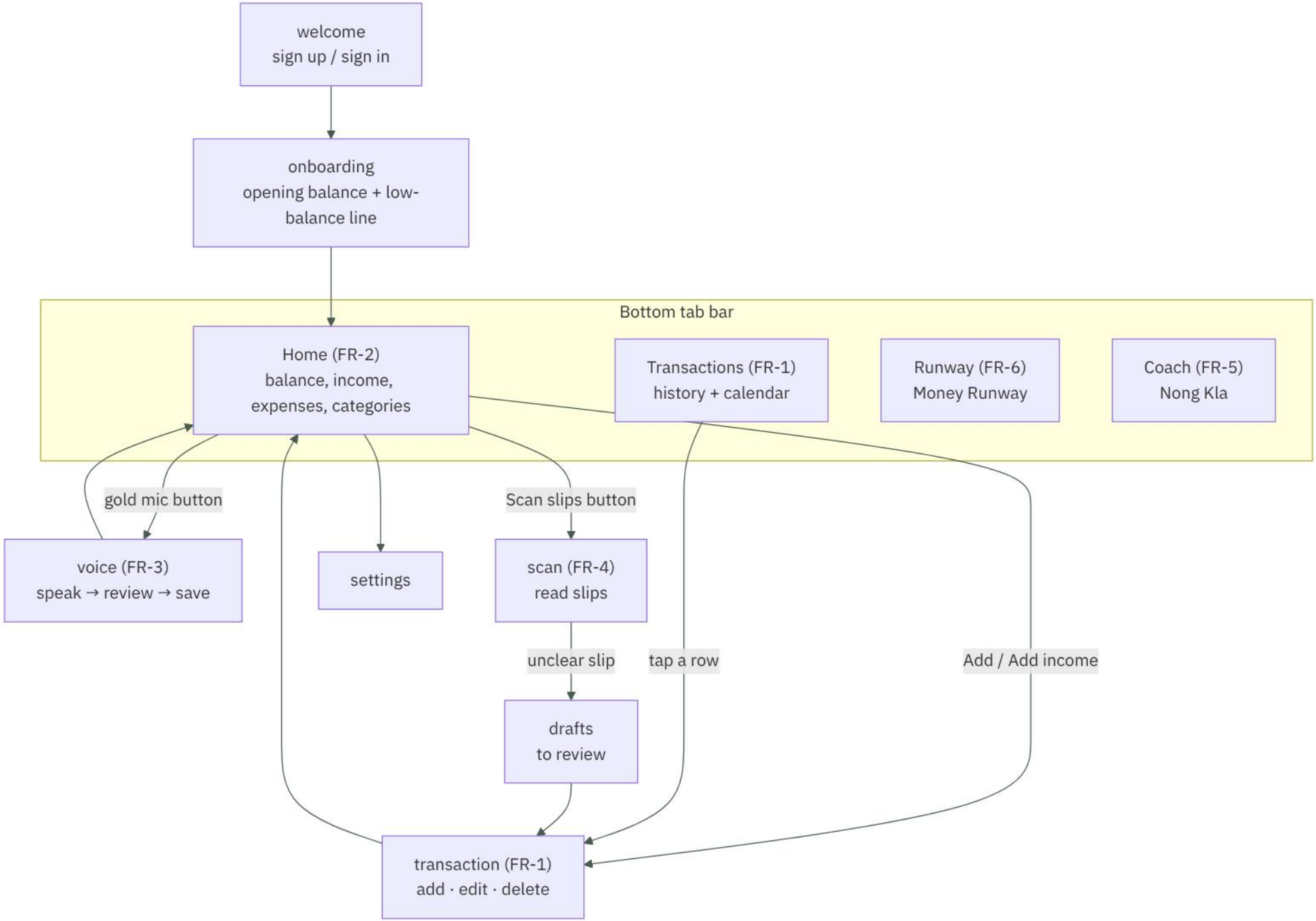
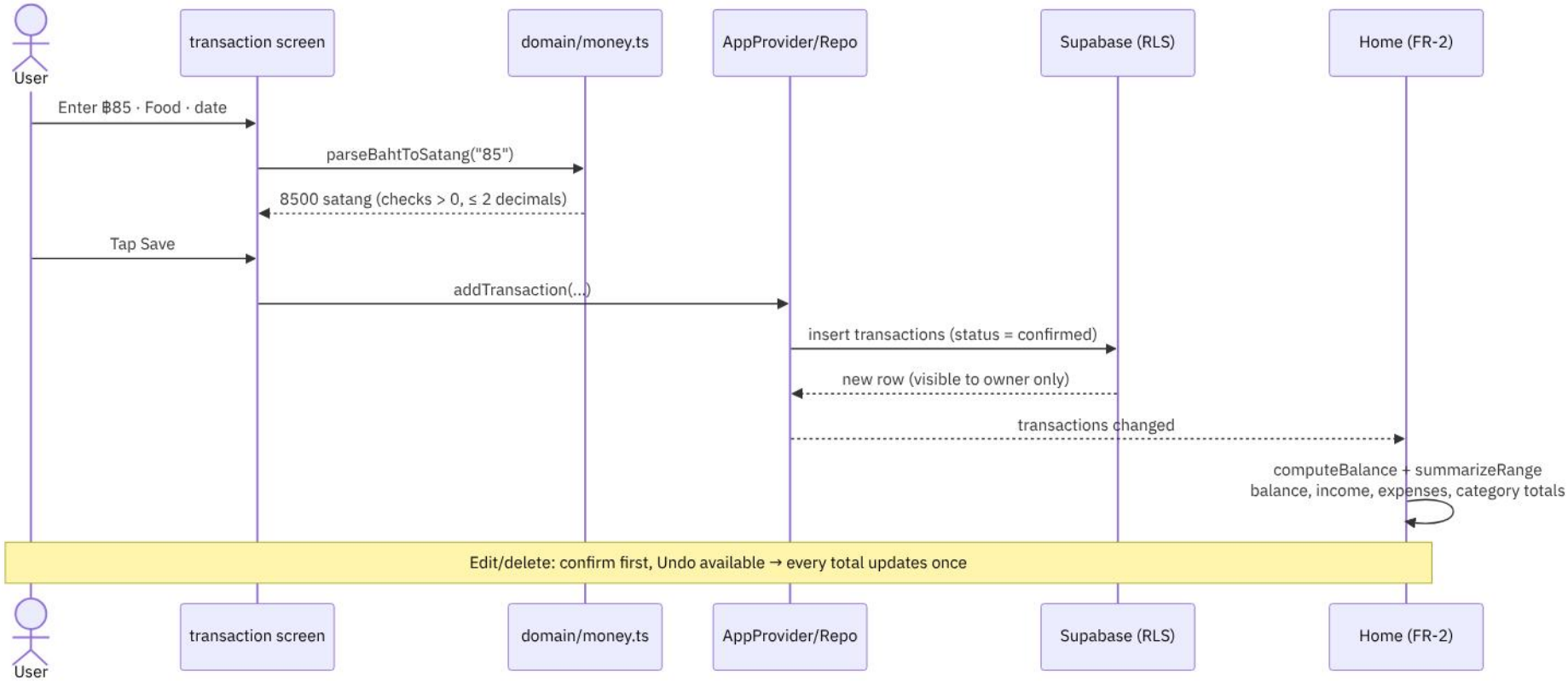
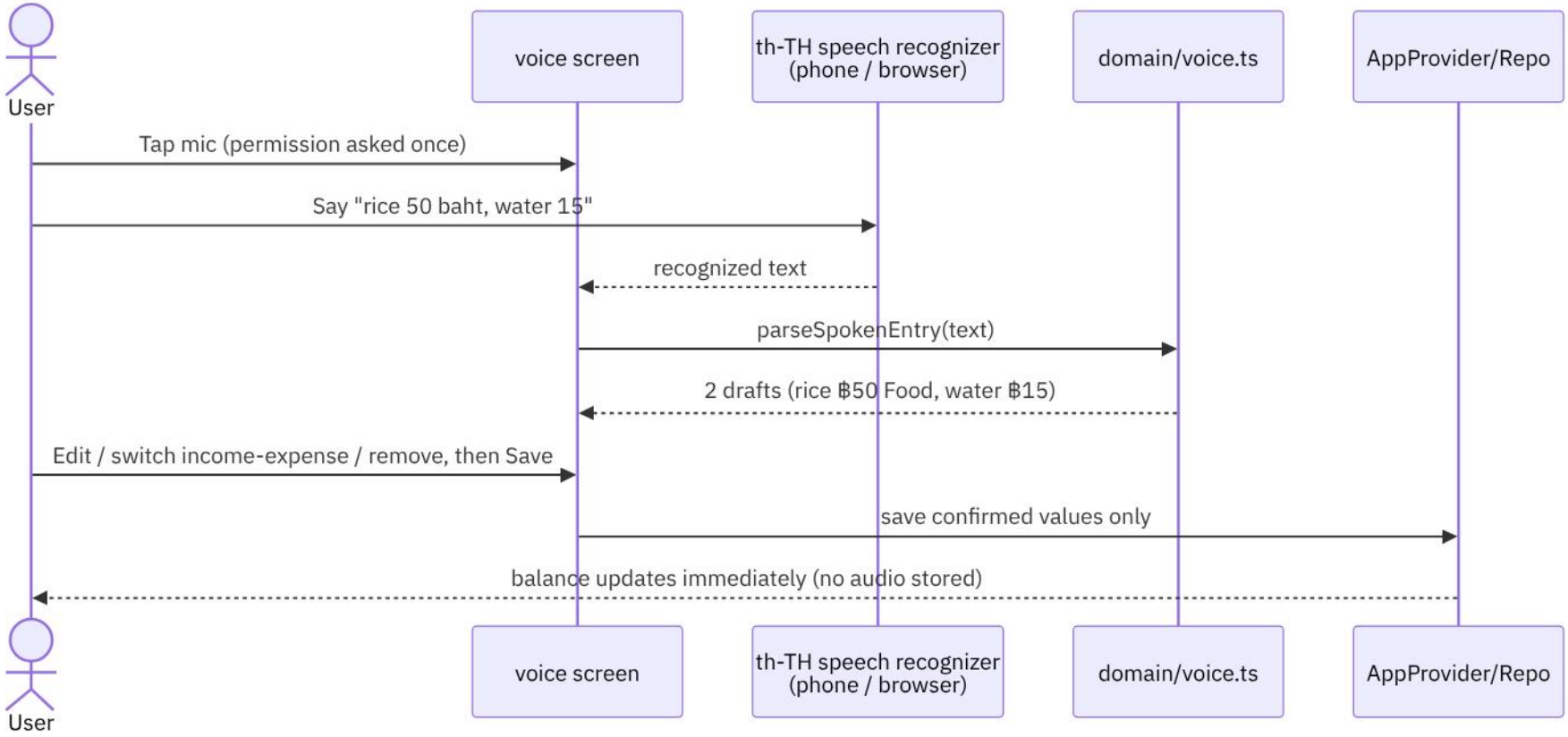

MindPay
Sign up / sign in
UC-2 View overview dashboard (FR-2)
UC-1 Manage transactions add, edit, categorize, delete (FR-1)
include
User (student / first jobber)
UC-3 Log a transaction by voice (FR-3)
include
UC-4 Scan gallery for slips (FR-4)
UC-5 Talk to AI coach Nong Kla (FR-5)
UC-6 View Money Runway (FR-6)
Phone / browser speech recognizer
AI service (Gemini via Edge Function)

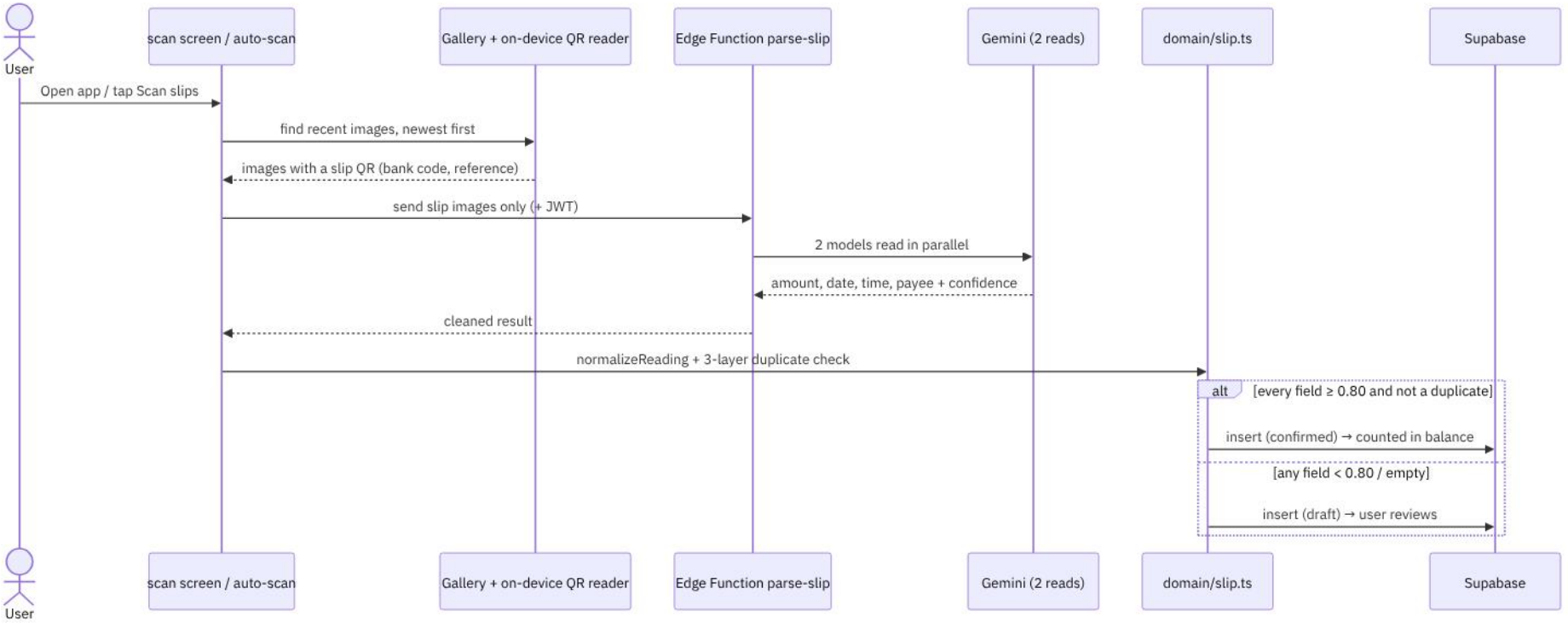

Phone / web app (React Native)
th-TH speech input
(phone / browser)
Rules and calculations
src/domain (unit tested)
Screens
src/app
AppProvider + Repo
src/data
Gallery scan
src/services
On-device slip QR reader
publishable key + RLS
Supabase PostgreSQL
profiles · transactions
savings_goals · ai_usage
Supabase Auth
email + password
slip images only
category totals only
Edge Function
parse-slip
Edge Function
coach
Google Gemini API
(key in Supabase Secrets)

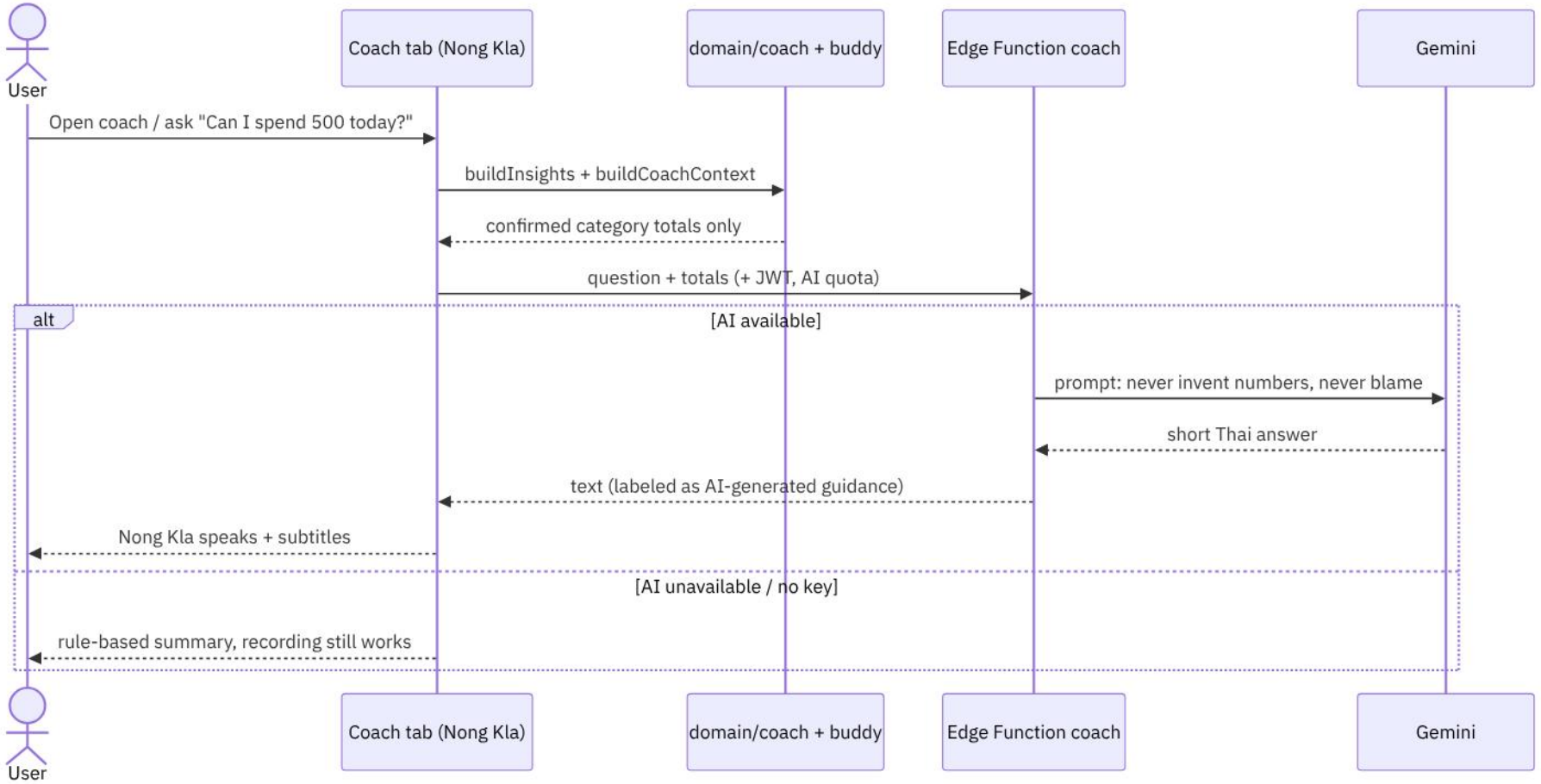

AUTH_USERS
has one
owns
sets
uses AI quota
PROFILES
uuid id PK
text display_name
bigint opening_balance_satang
bigint runway_floor_satang FR-6 low-balance line
bigint monthly_budget_satang
boolean onboarded
TRANSACTIONS
uuid id PK
uuid user_id FK
text kind income | expense
bigint amount_satang > 0
text category_key FR-1
text title
timestamptz occurred_at
text source manual | slip
text status draft | confirmed
text slip_ref FR-4 duplicate key
text slip_image_hash FR-4 duplicate key
numeric ocr_confidence FR-4 >= 0.80
SAVINGS_GOALS
uuid id PK
uuid user_id FK
text title
bigint target_satang
bigint saved_satang
date due_day
AI_USAGE
bigint id PK
uuid user_id FK
text kind slip | coach
timestamptz created_at

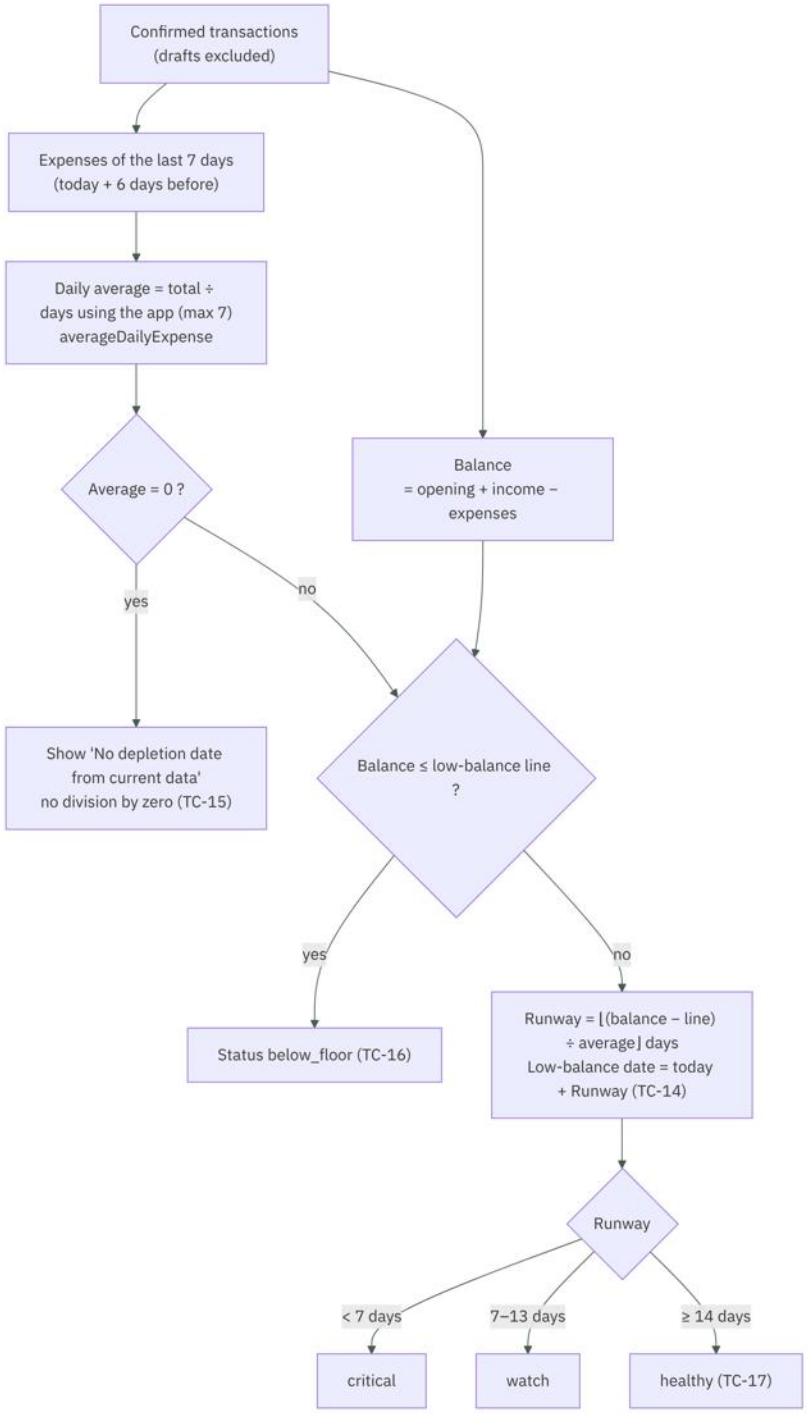

welcome
sign up / sign in
onboarding
opening balance + low-balance line
Bottom tab bar
Home (FR-2)
balance, income, expenses, categories
Transactions (FR-1)
history + calendar
Runway (FR-6)
Money Runway
Coach (FR-5)
Nong Kla
gold mic button
Scan slips button
voice (FR-3)
speak → review → save
settings
scan (FR-4)
read slips
unclear slip
tap a row
Add / Add income
drafts
to review
transaction (FR-1)
add · edit · delete


User
transaction screen
domain/money.ts
AppProvider/Repo
Supabase (RLS)
Home (FR-2)
Enter ฿85 · Food · date
parseBahtToSatang("85")
8500 satang (checks > 0, ≤ 2 decimals)
Tap Save
addTransaction(...)
insert transactions (status = confirmed)
new row (visible to owner only)
transactions changed
computeBalance + summarizeRange
balance, income, expenses, category totals
Edit/delete: confirm first, Undo available → every total updates once
User
transaction screen
domain/money.ts
AppProvider/Repo
Supabase (RLS)
Home (FR-2)


User
voice screen
th-TH speech recognizer (phone / browser)
domain/voice.ts
AppProvider/Repo
Tap mic (permission asked once)
Say "rice 50 baht, water 15"
recognized text
parseSpokenEntry(text)
2 drafts (rice ฿50 Food, water ฿15)
Edit / switch income-expense / remove, then Save
save confirmed values only
balance updates immediately (no audio stored)
User
voice screen
th-TH speech recognizer (phone / browser)
domain/voice.ts
AppProvider/Repo


User
scan screen / auto-scan
Gallery + on-device QR reader
Edge Function parse-slip
Gemini (2 reads)
domain/slip.ts
Supabase
Open app / tap Scan slips
find recent images, newest first
images with a slip QR (bank code, reference)
send slip images only (+ JWT)
2 models read in parallel
amount, date, time, payee + confidence
cleaned result
normalizeReading + 3-layer duplicate check
alt
[every field ≥ 0.80 and not a duplicate]
insert (confirmed) → counted in balance
[any field < 0.80 / empty]
insert (draft) → user reviews
User
scan screen / auto-scan
Gallery + on-device QR reader
Edge Function parse-slip
Gemini (2 reads)
domain/slip.ts
Supabase


User
Coach tab (Nong Kla)
domain/coach + buddy
Edge Function coach
Gemini
Open coach / ask "Can I spend 500 today?"
buildInsights + buildCoachContext
confirmed category totals only
question + totals (+ JWT, AI quota)
alt
[AI available]
prompt: never invent numbers, never blame
short Thai answer
text (labeled as AI-generated guidance)
Nong Kla speaks + subtitles
[AI unavailable / no key]
rule-based summary, recording still works
User
Coach tab (Nong Kla)
domain/coach + buddy
Edge Function coach
Gemini


Confirmed transactions
(drafts excluded)
Expenses of the last 7 days
(today + 6 days before)
Daily average = total ÷
days using the app (max 7)
averageDailyExpense
Average = 0 ?
yes
no
Show 'No depletion date
from current data'
no division by zero (TC-15)
Balance
= opening + income −
expenses
Balance ≤ low-balance line
?
yes
no
Status below_floor (TC-16)
Runway = ⌊(balance − line)
÷ average⌋ days
Low-balance date = today
+ Runway (TC-14)
Runway
< 7 days
7–13 days
≥ 14 days
critical
watch
healthy (TC-17)